KOMAL GUPTA

TOP 30 React INTERVIEW QUESTIONS

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1. **What is React?**
    a. React is a JavaScript library for building user interfaces, particularly single-page applications where data changes over time.
2. **What are the main features of React?**
    a. Virtual DOM, JSX, components, one-way data binding, and performance optimization.
3. **What is JSX?**
    a. JSX stands for JavaScript XML. It is a syntax extension for JavaScript that allows you to write HTML-like code inside JavaScript.
4. **What is the Virtual DOM?**
    a. The Virtual DOM is a lightweight copy of the actual DOM. React uses it to optimize rendering by updating only the parts of the DOM that have changed.

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5. **What are components in React?**
   - Components are the building blocks of a React application. They can be functional or class-based and can manage their own state.
6. **What is the difference between** functional and class components?
   - Functional components are stateless and written as functions. Class components can manage state and lifecycle methods, written as ES6 classes.
7. **What are props in React?**
   - Props (short for properties) are inputs to components. They are passed to components via HTML attributes and are immutable within the component.
8. **What is state in React?**
   - State is an object that holds data that may change over the lifetime of the component. It is managed within the component.

9. **What is the lifecycle of a React component**?
    - The lifecycle includes mounting, updating, and unmounting phases. Common lifecycle methods are `*componentDidMount*`, `*componentDidUpdate*`, and `*componentWillUnmount*`.
10. **How do you handle events in React?**
    - Events in React are handled using camelCase syntax and passing a function as the event handler, e.g., `*<button onClick={handleClick}>*`.
11. **What are hooks in React?**
    - Hooks are functions that let you use state and other React features in functional components. Examples include `*useState*`, `*useEffect*`, `*useContext*`.
12. **What is useState?**
    - useState is a hook that lets you add state to functional components. It returns an array with the current state and a function to update it.

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13. What is useEffect?

- useEffect is a hook that performs side effects in functional components. It replaces lifecycle methods like `componentDidMount`, `componentDidUpdate`, and `componentWillUnmount`.

14. What is the Context API?

- The Context API is a way to pass data through the component tree without having to pass props down manually at every level.

15. What is Redux?

- Redux is a state management library for JavaScript apps, often used with React to manage application state.

16. What is the difference between Redux and the Context API?

- Redux is a more complex state management solution with a larger ecosystem and more features, while the Context API is a simpler solution built into React for passing data.

17.What are higher-order components (HOCs)?

- HOCs are functions that take a component and return a new component, adding additional props or behavior.

18.What are controlled components?

- Controlled components are form elements that derive their values from the component's state, making the React component the single source of truth.

19.What are uncontrolled components?

- Uncontrolled components manage their own state internally. You use refs to access their current values.

20.What are refs in React?

- Refs provide a way to access the DOM nodes or React elements directly. They are used for managing focus, text selection, or triggering animations.

21. What is React Router?

- React Router is a library for routing in React applications, allowing you to handle navigation and rendering of different components based on the URL.

22. What is the significance of keys in React?

- Keys help React identify which items have changed, are added, or are removed. They should be unique among siblings to optimize rendering.

23. What is PropTypes in React?

- PropTypes is a typechecking library that allows you to check the types of props passed to a component to catch bugs and improve code readability.

24. What are fragments in React?

- Fragments let you group a list of children without adding extra nodes to the DOM, using <React.Fragment> or the shorthand syntax <>.

25.What is the difference between componentDidMount and useEffect?

- componentDidMount is a lifecycle method for class components, while useEffect is a hook for functional components that can perform side effects.

26.How do you optimize performance in a React application?

- Use techniques like code splitting, lazy loading, memoization (React.memo), avoiding unnecessary re-renders, and using the Virtual DOM efficiently.

27.What is code splitting in React?

- Code splitting is a technique to split your code into smaller bundles that can be loaded on demand, improving load times and performance.

28.What is lazy loading in React?

- Lazy loading is a technique to load components only when they are needed, using React.lazy and Suspense for dynamic imports.

29.What are error boundaries in React?

- Error boundaries are components that catch JavaScript errors in their child component tree, logging the errors and displaying a fallback UI.

30.What is the difference between React.memo and React.PureComponent?

- React.memo is used to optimize functional components by memoizing their output. React.PureComponent does the same for class components by implementing a shallow comparison on props and state.

These questions cover a range of fundamental and advanced topics that are frequently asked in React interviews. Preparing answers for these will help you demonstrate a solid understanding of React principles and practices.

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